

# Lab 5: Differential Amplifier

ECE 0102: Microelectronic Circuits + Lab

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## Team Roles:

| Activity             | Student Name             |
|----------------------|--------------------------|
| Circuit Construction | Stefan Zhang             |
| Data Collection      | Knight Liu               |
| Data Analysis        | Stefan Zhang; Knight Liu |
| Report Writing       | Stefan Zhang             |

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## 1) Objectives

The objective of this lab is to construct and evaluate a differential amplifier using an LM741 operational amplifier. The differential-mode gain, common-mode gain, and common-mode rejection ratio (CMRR) are measured and calculated.

## 2) Equipment Used

**Table 1:** Equipment used in Lab 5

| Item                               | Quantity |
|------------------------------------|----------|
| LM741CN/NOPB operational amplifier | 1        |
| 1 k $\Omega$ resistor              | 2        |
| 10 k $\Omega$ resistor             | 2        |
| Function generator                 | 1        |
| Oscilloscope                       | 1        |
| DC power supply                    | 1 set    |
| Breadboard and wires               | 1 set    |

### 3) Action 1.1: Differential Input

For the ideal op-amp, the non-inverting input voltage is

$$V_+ = \frac{R_4}{R_3 + R_4} V_2$$

Applying KCL at the inverting input node,

$$\frac{V_1 - V_-}{R_1} + \frac{V_{OUT} - V_-}{R_2} = 0$$

Therefore,

$$V_{OUT} = \left(1 + \frac{R_2}{R_1}\right) \frac{R_4}{R_3 + R_4} V_2 - \frac{R_2}{R_1} V_1$$

When  $R_1 = R_3$  and  $R_2 = R_4$ ,

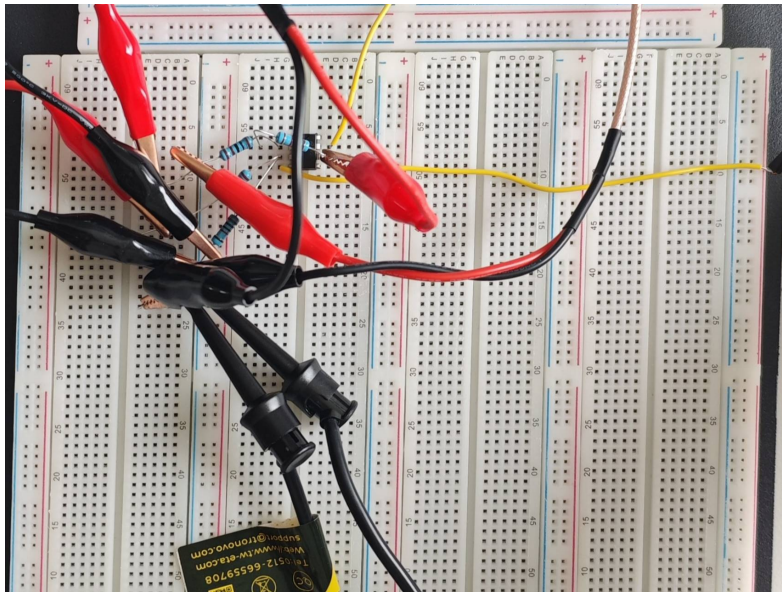
$$V_{OUT} = \frac{R_2}{R_1} (V_2 - V_1)$$

In this lab,  $R_1 = R_3 = 1 \text{ k}\Omega$ ,  $R_2 = R_4 = 10 \text{ k}\Omega$ . Thus,

$$V_{OUT} = 10(V_2 - V_1)$$

For the differential input,  $V_2(t) = -V_1(t)$ , so

$$V_{OUT}(t) = -20V_1(t)$$



**Figure 1:** Circuit connection for Action 1.1



**Figure 2:** Oscilloscope capture of  $V_1(t)$  and  $V_{OUT}(t)$  for Action 1.1

**Table 2:** Measured data for Action 1.1

| Quantity         | Measured Value |
|------------------|----------------|
| $V_{1,pp}$       | 1.040 V        |
| $V_{2,pp}$       | 1.040 V        |
| $V_{OUT,pp}$     | 14.20 V        |
| Frequency        | 1.00 kHz       |
| Phase difference | 180°           |

#### 4) Action 1.2: Common-Mode Input

For common-mode input,

$$V_1 = V_2 = V_{cm}$$

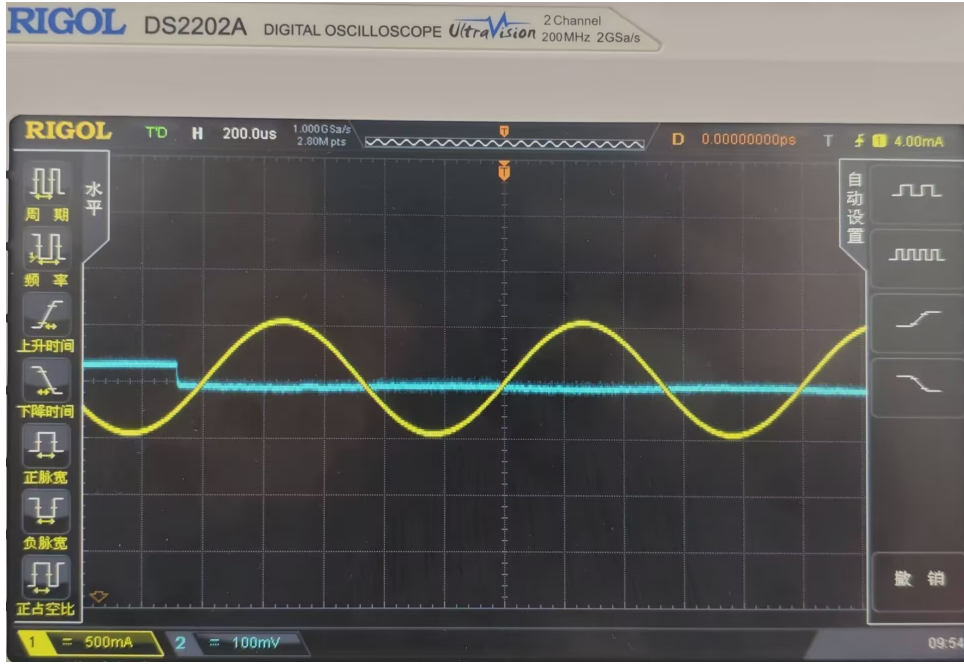
Since

$$V_{OUT} = 10(V_2 - V_1)$$

the ideal output is

$$V_{OUT} = 10(V_{cm} - V_{cm}) = 0$$

Thus, the ideal common-mode gain is zero.



**Figure 3:** Oscilloscope capture of  $V_1(t)$  and  $V_{OUT}(t)$  for Action 1.2

**Table 3:** Measured data for Action 1.2

| Quantity         | Measured Value |
|------------------|----------------|
| $V_{cm,pp}$      | 1.080 V        |
| $V_{OUT,pp}$     | 28.00 mV       |
| Frequency        | 1.00 kHz       |
| Phase difference | 0°             |

### 5) Action 1.3: Gain and CMRR

The differential-mode input gain is

$$A_d = \frac{V_{OUT,pp}}{V_{d,pp}}.$$

For Action 1.1, the two inputs have equal amplitude and opposite phase, so

$$V_{d,pp} = 2V_{1,pp} = 2(1.040 \text{ V}) = 2.080 \text{ V}$$

Therefore,

$$A_d = \frac{14.20 \text{ V}}{2.080 \text{ V}} = 6.83$$

The theoretical value is

$$A_{d,theory} = 10$$

The common-mode input gain is

$$A_{cm} = \frac{V_{OUT,pp}}{V_{cm,pp}}$$

Using the measured values,

$$A_{cm} = \frac{28.00 \text{ mV}}{1.080 \text{ V}} = \frac{0.0280 \text{ V}}{1.080 \text{ V}} = 0.0259.$$

The common-mode rejection ratio is

$$CMRR = \left| \frac{A_d}{A_{cm}} \right|$$

Thus,

$$CMRR = \left| \frac{6.83}{0.0259} \right| = 263.7$$

In decibels,

$$CMRR_{dB} = 20 \log_{10}(263.7) = 48.4 \text{ dB}$$

**Table 4:** Calculated gain and CMRR

| Quantity                     | Formula                             | Result  |
|------------------------------|-------------------------------------|---------|
| Differential-mode gain $A_d$ | $\frac{V_{OUT,pp}}{2V_{1,pp}}$      | 6.83    |
| Common-mode gain $A_{cm}$    | $\frac{V_{OUT,pp}}{V_{cm,pp}}$      | 0.0259  |
| CMRR                         | $\left  \frac{A_d}{A_{cm}} \right $ | 263.7   |
| CMRR in dB                   | $20 \log_{10}(CMRR)$                | 48.4 dB |

## 6) Discussion

The theoretical output of the differential amplifier is

$$V_{OUT} = 10(V_2 - V_1)$$

In Action 1.1, the two input signals have equal amplitude and opposite phase, so the measured differential input is

$$V_{d,pp} = 2.080 \text{ V}$$

The measured output is

$$V_{OUT,pp} = 14.20 \text{ V}$$

so the measured differential-mode gain is

$$A_d = 6.83$$

This is lower than the theoretical gain of 10. The ideal output should be approximately

$$V_{OUT,pp} = 10(2.080 \text{ V}) = 20.8 \text{ V}$$

However, the LM741 is powered by only  $\pm 12 \text{ V}$ , and its output cannot swing exactly to the supply rails. Therefore, the output waveform is limited by saturation, which reduces the measured peak-to-peak output voltage.

In Action 1.2, the same signal is applied to both inputs. Ideally, the output should be zero. The measured common-mode output is

$$28.00 \text{ mV}_{pp}$$

which gives

$$A_{cm} = 0.0259$$

The small nonzero output is mainly caused by resistor mismatch, finite CMRR of the LM741, offset voltage, and measurement noise.

The measured CMRR is

$$CMRR = 263.7$$

This means that the circuit rejects common-mode signals much more strongly than it amplifies them.

## 7) Conclusion

In this lab, a differential amplifier was constructed using an LM741 operational amplifier. The output expression was derived as

$$V_{OUT} = \left(1 + \frac{R_2}{R_1}\right) \frac{R_4}{R_3 + R_4} V_2 - \frac{R_2}{R_1} V_1.$$

For the matched resistor condition  $R_1 = R_3$  and  $R_2 = R_4$ , this equation simplifies to

$$V_{OUT} = \frac{R_2}{R_1}(V_2 - V_1).$$

With  $R_1 = R_3 = 1\text{ k}\Omega$  and  $R_2 = R_4 = 10\text{ k}\Omega$ , the output becomes

$$V_{OUT} = 10(V_2 - V_1).$$

The theoretical differential-mode gain is 10, while the ideal common-mode gain is 0. Therefore, the ideal CMRR is infinite. The measured CMRR is expected to be finite because the physical circuit contains resistor mismatch and nonideal op-amp effects.

## References

- [1] SCUPI 2026S Microelectronics Circuit Lab Manual, Lab #5: Differential Amplifier.
- [2] Sedra, A. S., Smith, K. C., Carusone, T. C., and Gaudet, V., *Microelectronic Circuits*, Oxford University Press.
- [3] Texas Instruments, *LM741 Operational Amplifier Datasheet*.